Batch and stream

Data engineering · Level 4 · Lesson 4



In this lesson you will

- Explain batch and stream processing and their trade-offs.
- Tame noise with a moving window, and see the delay it adds.
- Choose batch, stream or both for real systems.

Key ideas

- Batches are simple and late; streams are fast and always on.
- Bigger windows: fewer false alarms, later alerts.
- Use event time, and expect duplicates.

Human stream

1. Readings (3 min): one a minute
2. Window 1 (3 min): alarm on every reading
3. Window 3 (5 min): average the last three
4. Decide (4 min): batch, stream or both?

Use the readings from the Python step, so the class can check its alerts against the code.

Let's discuss

1. Why is a batch easy to rerun?
2. What did the bigger window cost us?
3. Why use the time an event happened, not when it arrived?

Exit ticket

- Name one system that needs a stream and one that a batch serves better, with a reason each.